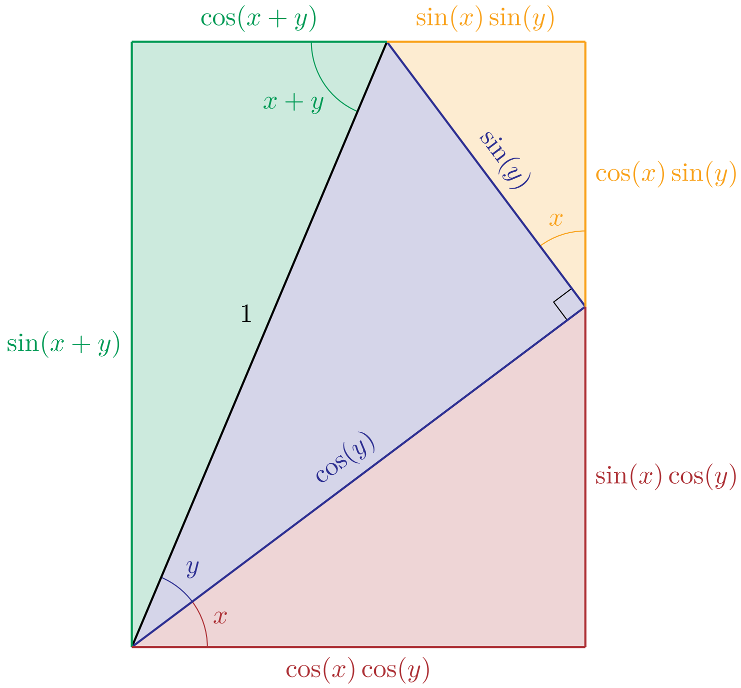
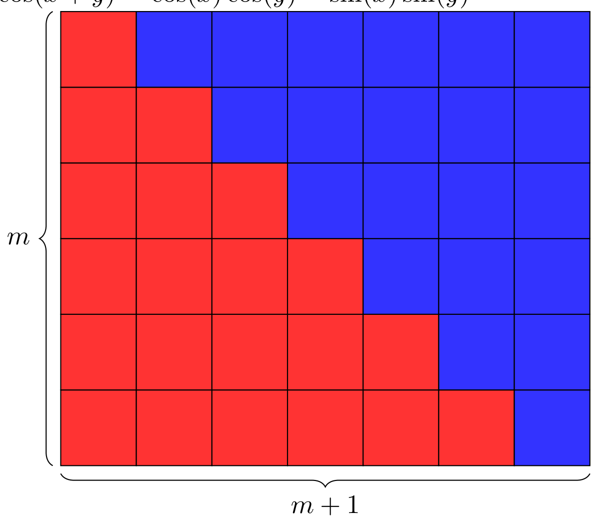
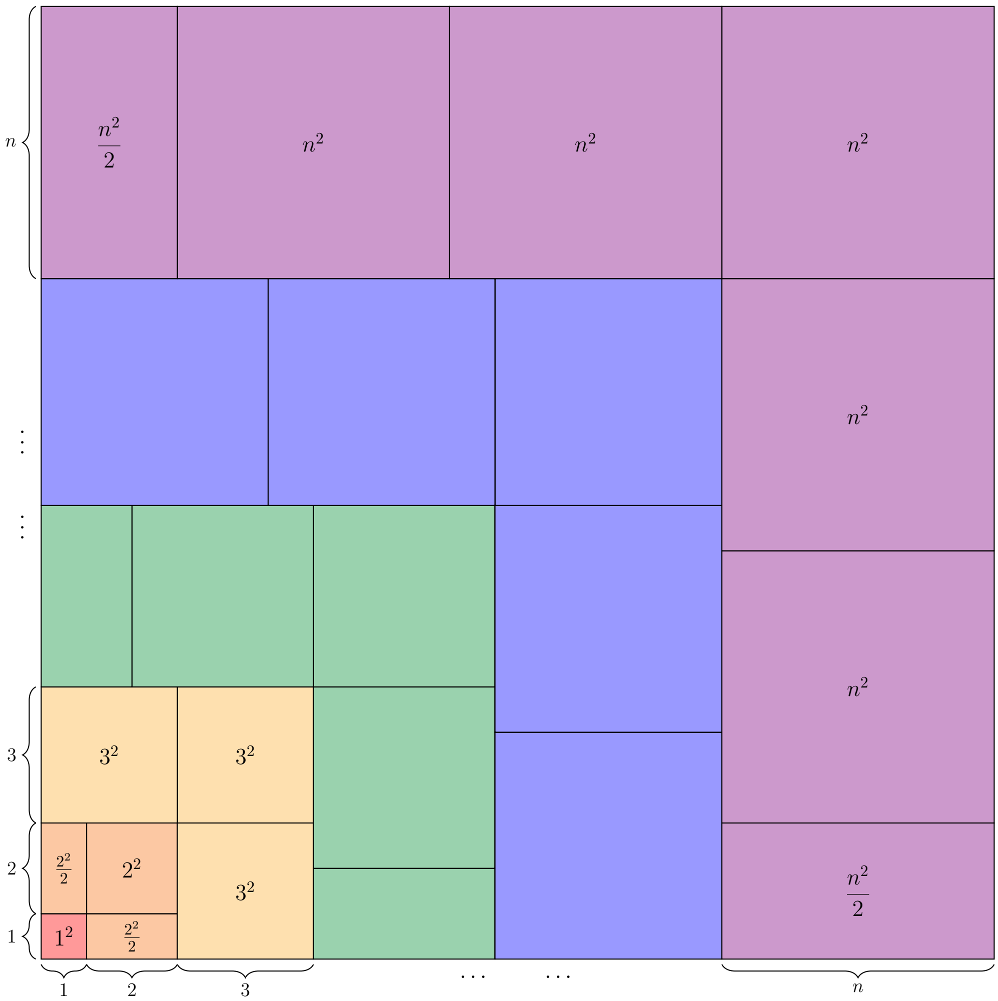
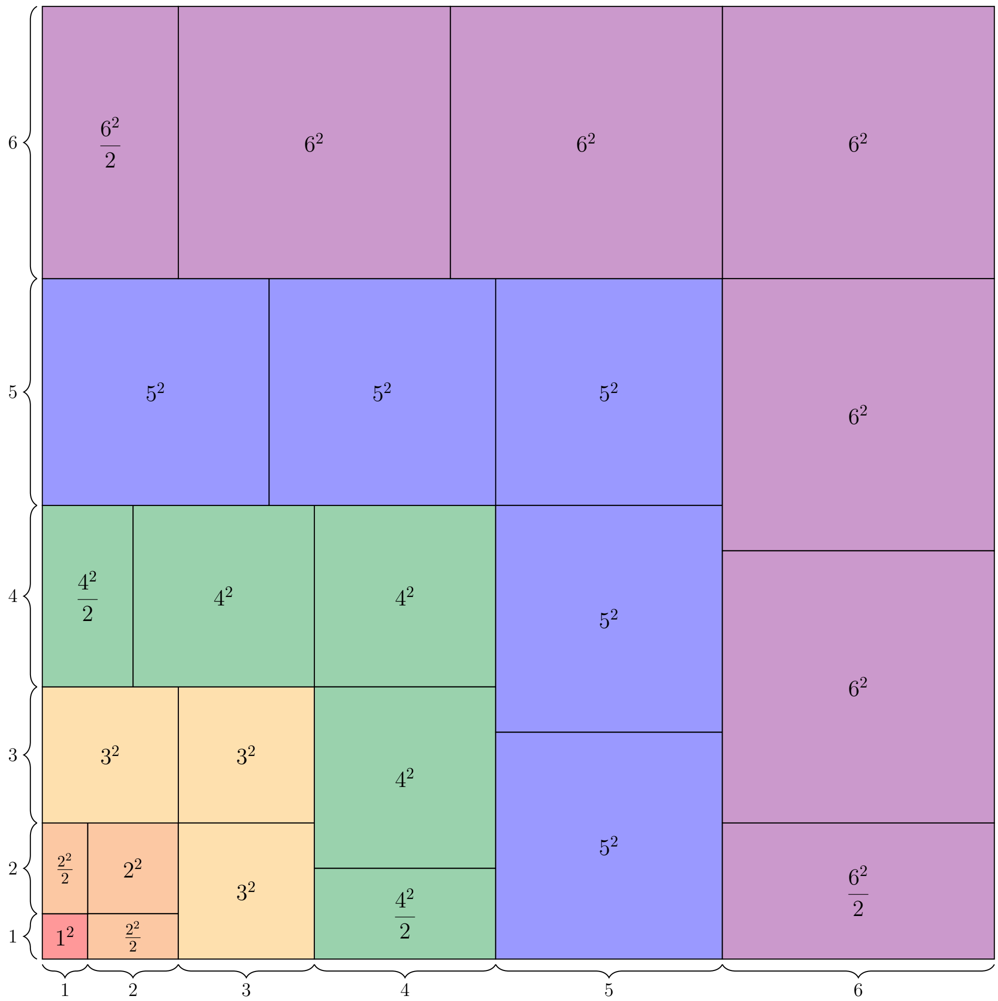


$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y)$
$\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$




n
$\frac{n^2}{2}$
n^2
n^2
n^2
n^2
n^2
$\frac{n^2}{2}$
$\vdots$
$\vdots$
3
3^2
3^2
3^2
2
$\frac{2^2}{2}$
2^2
1
1^2
$\frac{2^2}{2}$
1
2
3
$\cdots$
$\cdots$
n



Theorem -1.1 (Hockey Stick Identity)**.** For all $n, k \in \mathbb{N}_0$, we have:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$$

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

1 7 21 35 35 21 7 1

1 8 28 56 70 56 28 8 1

1 9 36 84 126 126 84 36 9 1

1 10 45 120 210 252 210 120 45 10 1

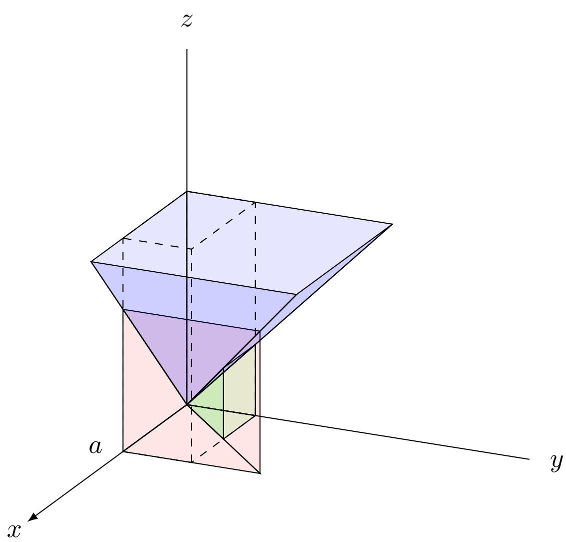
1 11 55 165 330 462 462 330 165 55 11 1

1 12 66 220 495 792 924 792 495 220 66 12 1

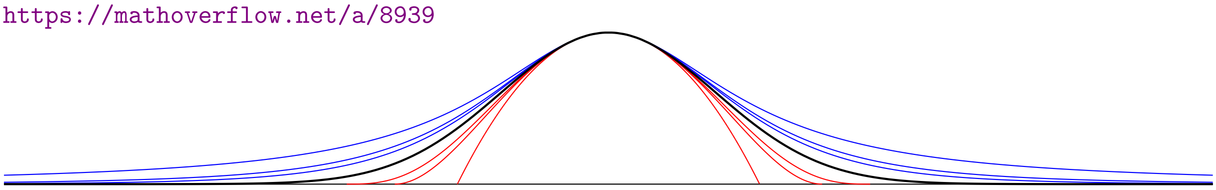
1 13 78 286 715 1287 1716 1716 1287 715 286 78 13 1

1 14 91 364 1001 2002 3003 3432 3003 2002 1001 364 91 14 1

$$\frac{a^3+b^3+c^3}{3} \geq abc$$

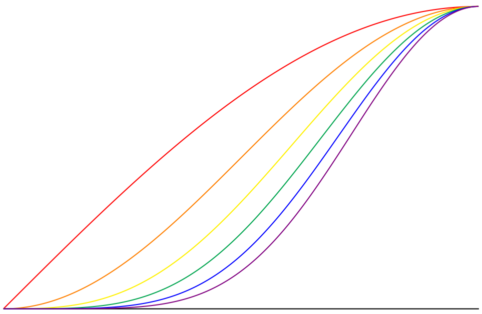


https://mathoverflow.net/a/8939



$$\int_{-\sqrt{n}}^{\sqrt{n}} \left(1-\frac{x^2}{n}\right)^n dx = \sqrt{n}\int_{-1}^{1}\left(1-y^2\right)^n dy \qquad (x=\sqrt{n}y)$$
$$= \sqrt{n}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t)^{2n+1}\, dt \qquad (y=\sin(t))$$

$$\int_{-\infty}^{\infty} \left(1+\frac{x^2}{n}\right)^{-n} dx = \sqrt{n}\int_{-\infty}^{\infty}\left(1+y^2\right)^{-n} dy \qquad (x=\sqrt{n}y)$$
$$= \sqrt{n}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t)^{2n-2}\, dt \qquad (y=\tan(t))$$



$$\int_0^{\frac{\pi}{2}} \sin(x)^{2m}\, dx = \frac{(2m)!}{2^{2m+1}(m!)^2}\pi$$
$$\int_0^{\frac{\pi}{2}} \sin(x)^{2m+1}\, dx = \frac{2^{2m}(m!)^2}{(2m+1)!}$$

Comparing $2m$ with $2m+1$:

$$\frac{(2m)!}{2^{2m+1}(m!)^2}\pi > \frac{2^{2m}(m!)^2}{(2m+1)!}$$
$$\pi > \frac{2^{4m+1}(m!)^4}{(2m)!(2m+1)!}$$

Comparing $2m$ with $2m-1$:

$$\frac{(2m)!}{2^{2m+1}(m!)^2}\pi < \frac{2^{2m-2}((m-1)!)^2}{(2m-1)!}$$
$$\pi < \frac{2^{4m-1}(m!)^2((m-1)!)^2}{(2m)!(2m-1)!}$$
$$\frac{}{(2m-1)!}\frac{(2m+1)!}{4m^2} > 1$$

$$\begin{array}{cccccccc}
1 & r & r^2 & r^3 & r^4 & r^5 & \cdots \\
r & r^2 & r^3 & r^4 & r^5 & \ddots \\
r^2 & r^3 & r^4 & r^5 & \ddots \\
r^3 & r^4 & r^5 & \ddots \\
r^4 & r^5 & \ddots \\
r^5 & \ddots \\
\vdots
\end{array}$$